Ghent University
Faculty of Economics and Business Administration
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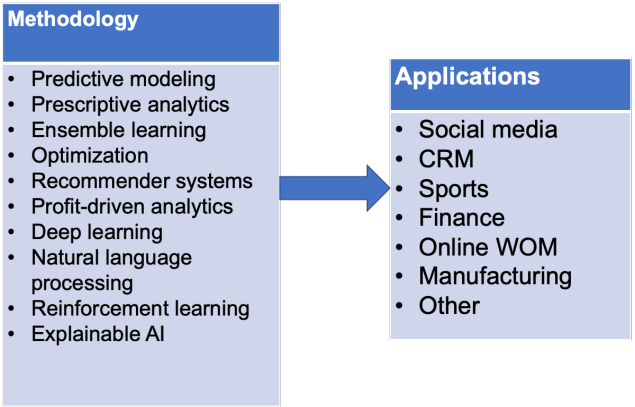
Matthias Bogaert (PhD)

Associate Professor of Data Analytics

Summary

Matthias Bogaert is an Assistant Professor of Data Analytics at the Marketing, Innovation, and Organization Department (Research Group Data Analytics) at Ghent University. He is also a member of the Flanders Make at UGent corelab CVAMO. Prior to joining Ghent University, he worked as a lecturer in Business Analytics at the University of Edinburgh Business School. He has taught a wide-range of data analytics courses ranging from basic statistics and database management to advanced predictive analytics and social media and web analytics. His research focuses on applications of descriptive, predictive, explainable and prescriptive analytics in social media, digital communities, sports, customer relationship management, and production and manufacturing. The main goal of his research is to come up with a reproducible data analytical methodology that companies can easily implement.



Education

PhD in Business Economics

2014 – 2018 Ghent University

Title doctoral dissertation: Harnessing the Power of Social Media in Predictive Analytics.
Supervisors: Prof. Dr Dirk Van den Poel and Prof. Dr Michel Ballings
Examination Board: Prof. Dr Patrick Van Kenhove, Prof. Dr Dries Benoit, Prof. Dr Bart Larivière, Prof. Dr Bart Baesens, Prof. Dr Koen De Bock

Master of Science in Applied Economics: Business Engineering, Marketing Engineering /Data Analytics

2012 – 2014 Ghent University

Grade: Magna cum laude (800/1000)

Bachelor of Science in Applied Economics: Business Engineering
2009 – 2012 Ghent University

Grade: cum laude (746/1000)

Employment

Associate Professor in Data Analytics at Ghent University (100%)
September 2025 – Present

Assistant Professor in Data Analytics at Ghent University (100%)
September 2020 – Present

Visiting Professor in Business Analytics and Big Data at Université de Namur (5%)
February 2020 – Present

Doctor-assistant at KU Leuven (100%)
September 2019 – September 2020

Visiting Professor in Social Media and Web Analytics at Ghent University (10%)
January 2019 – September 2020

Lecturer in Business Analytics at the University of Edinburgh Business School (100%)
September 2018 – September 2019

PhD researcher and teaching assistant at Ghent University (100%)
September 2014 – September 2018
Supervisor Prof. Dr Dirk Van den Poel, co-supervisor: Prof. Dr Michel Ballings

Journal Articles

1. Janssens, W., Bogaert, M., & Van den Poel, D. (2025). *A comparative analysis of topic reduction techniques for BERTopic*. IEEE Access, 13, 204087-204103.
2. De Bock, K. W., Bogaert, M., & du Jardin, P. (2025). *Ensemble learning for operations research and business analytics*. Annals of Operations Research, 353, 419–448.
3. Ballegeer, M., Bogaert, M., & Benoit, D. F. (2025). *Evaluating the stability of model explanations in instance-dependent cost-sensitive credit scoring.* European Journal of Operational Research, *326*(3), 630-640.
4. Wang, X., Janssens, B., Bogaert, M., Vanderbauwhede, L., & Schetgen, L. (2025). *Determining well-being during a crisis based on Twitter.* Annals of Operations Research, 1-38.
5. Rahman, S., Janssens, B., & Bogaert, M. (2025). *Profit-driven pre-processing in B2B customer churn modeling using fairness techniques.* Journal of Business Research, *189*, 115159.
6. Janssens, B., & Bogaert, M. (2025). *PerfoRank: cluster-based performance ranking for improved performance evaluation and estimation in professional cycling*. Machine Learning, 114(1), 1-30.
7. Larivière, B., Schetgen, L., Bogaert, M., & Van den Poel, D. (2024). *Customer Experiences and Coping Behaviors during Crisis Situations: The Role of Service Adaptation and Service Transformation*. Journal of Business Research, 188, 115089.
8. Janssens, B., Schetgen, L., Bogaert, M., Meire, M., & Van den Poel, D. (2024). *360 Degrees rumor detection: When explanations got some explaining to do*. European Journal of Operational Research, *317*(2), 366-381.
9. Janssens, B., Bogaert, M., Bagué, A., & Van den Poel, D. (2024). *B2Boost: Instance-dependent profit-driven modelling of B2B churn*. Annals of Operations Research, *341*(1), 267-293.
10. Bogaert, M., & Delaere, L. (2023). *Ensemble methods in customer churn prediction: A comparative analysis of the state-of-the-art.* Mathematics, 11(5), 1137.

11. Janssens, B., Bogaert, M. & Maton, M. (2022). *Predicting the next Pogačar: a data analytical approach to detect young professional cycling talents.* Annals of Operations Research, 325(1), 557-588.
12. Janssens, B., Bogaert, M., & Van den Poel, D. (2021). *Evaluating the influence of Airbnb listings' descriptions on demand.* International Journal of Hospitality Management, 99, 103071.
13. Bogaert, M., Ballings, M., Van den Poel, D., & Oztekin, A. (2021). *Box office sales and social media: A cross-platform comparison of predictive ability and mechanisms.* Decision Support Systems, 147, 113517.
14. Schetgen, L., Bogaert, M., & Van den Poel, D. (2021). *Predicting donation behavior: Acquisition modeling in the nonprofit sector using Facebook data.* Decision Support Systems 141, 133446.
15. Bogaert, M., Ballings, M., Bergmans, R., & Van den Poel, D. (2021). *Predicting Self-declared Movie Watching Behavior Using Facebook Data and Information-Fusion Sensitivity Analysis.* Decision Sciences, 52(3), 776–810.
16. Bogaert, M., Lootens, J., Van den Poel, D., Ballings, M., 2019. *Evaluating Multi-label Classifiers and Recommender Systems in the Financial Service Sector.* European Journal of Operational Research 279 (2), 620-634.
17. Bogaert, M. 2018. *Harnessing the Power of Social Media in Predictive Analytics.* Doctoral Dissertation UGent.
18. Bogaert, M., Ballings, M., Van den Poel, D., 2018. *Evaluating the Importance of Different Communication Types in Romantic Tie Prediction on Social Media.* Annals of Operations Research 263, 501-527.
19. Bogaert, M., Ballings, M., Hosten, M., Van den Poel, D., 2017. *Identifying Soccer Players on Facebook Through Predictive Analytics.* Decision Analysis 14 (4), 274 -297.
20. Bogaert, M., Ballings, M., Van den Poel, D., 2016. *The Added Value of Facebook Friends Data in Event Attendance Prediction.* Decision Support Systems 82, 26–34.
21. Ballings, M., Van den Poel, D., Bogaert, M., 2016. *Social Media Optimization: Identifying an Optimal Strategy for Increasing Network Size on Facebook.* Omega – International Journal of Management Science, Business Analytics 59, Part A, 15–25.

Conference Proceedings

1. Brito, Heldt, Silveira, C., Bogaert, M., Bucco, G. B., Luce, F. B., Becker, J. L., Zabala, F., & Anzanello, M. J. (2024). *Predicting and Explaining Customer Data Sharing in the Open Banking.* MIDAS 2024 : 9th Workshop on MIning DAta for Financial applicationS. European Conference on Machine Learning and Principles and Practice of Knowledge Discovery, Vilnius, Lithuania.
2. Janssens, B., Bogaert, M., & Verstockt, S. (2024). *Large Language Models on Race Commentary: Towards Granular Data in Cycling Analytics.* MLSA 2025: 11th Workshop on Machine Learning and Data Mining for Sports Analytics. European Conference on Machine Learning and Principles and Practice of Knowledge Discovery, Vilnius, Lithuania.
3. Janssens, B., & Bogaert, M. (2023). *Performance Measurement 2.0: Towards a Data-Driven Cyclist Specialization Evaluation.* In International Workshop on Machine Learning and Data Mining for Sports Analytics (pp. 179-190). Cham: Springer Nature Switzerland.
4. Baron, E., Janssens, B., & Bogaert, M. (2023). *Bike2Vec: vector embedding representations of road cycling riders and races.* Proceedings of the 10th MathSport International Conference. arXiv preprint arXiv:2305.10471.
5. Janssens, B., & Bogaert, M. (2022). *Imputation of Non-participated Race Results.* In International Workshop on Machine Learning and Data Mining for Sports Analytics (pp. 155-166). Springer, Cham.

Awards

Winner Lazaridis 2021 Most Promising Research Award

November 2021

Paper: Text mining customer experience with covid-19 events: A longitudinal and cross-industry analysis of the impact on customer well-being.

Runner-up INFORMS Social Media Analytics Best Student Paper Award

November 2021
Paper: Developing a detection and interpretation tool for rumors regarding COVID-19 on Twitter using deep learning.

Runner-up INFORMS Social Media Analytics Best Student Paper Award

November 2016
Paper: Evaluating the Importance of Different Communication Types in Romantic Tie Prediction on Social Media

Academic Teaching

Data analytics (course organizer, 4ECTS)

Université de Namur, September 2025 – Present
Bsc in business engineering
Programming language: Python.

Data Visualization (6ECTS)

IESEG School of Management, January 2025 – Present
PGE: Master (Grande Ecole)
Tool: Tableau.

Predictive and prescriptive analytics, advanced predictive analytics (6ECTS)

Ghent University, February 2021 – Present
Msc in business engineering: data analytics and Msc in data science for business.
Programming language: Python.

Business analytics and big data (course organizer, 3ECTS)

Université de Namur, February 2020 – August 2020
Msc in business engineering: analytics and digital business.

Social media and web analytics (course organizer, 6ECTS)

Ghent University, February 2020 – Present
Msc in business engineering: data analytics and Msc in data science for business.
Programming language: Python.

Principles of Database Management (course organizer, 6ECTS)

KU Leuven, September 2019 – January 2020
Msc in Information Management and electives from Msc in business engineering and bioinformatics.
Programming language: SQL.

Social media and web analytics (course organizer, 6ECTS)

Ghent University, February 2019 - March 2019
Msc in business engineering: data analytics and Msc in marketing analysis.
Programming language: R.

Data mining (co-lecturer, 8ECTS)

University of Edinburgh, January 2019 – May 2019
Optional course in the Msc in marketing and business analysis.
Programming language: R.

Business research methods 1: introduction to data analysis (co-lecturer, 10ECTS)
University of Edinburgh, September 2018-December 2018
Undergraduate course in business studies. I gave the lectures on parametric and non-parametric testing.

Predictive analytics and modelling of data (course organizer, 7.5ECTS)
University of Edinburgh, September 2018 – December 2018
Msc in business analytics.
Programming language: R.

Predictive and prescriptive analytics, advanced predictive analytics (assistant, 6ECTS)
Ghent University, January 2015 – June 2018
Msc in business engineering: data analytics and Msc in marketing analysis.
Programming language: R.

Professional Teaching

FEB Academy (Day workshop)
Ghent University, May 2025
Session: Social Media Analytics

Postgraduate in Big Data and Analytics (3 sessions)
KU Leuven, November 2019 – present
Sessions: Text Mining and NLP, Social Media Analytics

KBC ADAM Bootcamp (2 sessions)
KBC Bank NV, 4-5 November 2019
Session 3: Modeling and Evaluation, Session 4: Piloting and Deployment

Project Grants

Strategic Basic Research (SBO) (€543,167 (partner core lab, total budget of €2,543,372))
October 2025– present.
Automated Generation and Lifecycle Cost Assessment of Manufacturing System Design under Multi-Product Demand Uncertainty (Ageas)

Industrial Research Valorization and Acceleration (IRVA) (€510,709 (leading core lab, total budget of €1,333,333))
March 2025– present.
Cost-sensitive learning in production for optimal maintenance strategies (COSTLEAP).

Research and Development in collaboration with Duco (€50,500)
January 2025– June 2025.
Active House (VLAIO O&O)

Strategic Basic Research (SBO) (€1,051,531 (leading core lab, total budget of €3,278,843))
June 2024 – present.
Prediction & Management of the 'Reverse' Remanufacturing Supply Chain (REMASC).

FWO Junior Postdoctoral Fellowship (€335,000)
September 2022 – present
Promotor of Dr. Bram Janssens: UniCycL: Uniform & General Framework for Textual Input Representation in Professional Cycling Talent Management.

FEB UGent Teaching Innovation Grant (€30,000)
September 2022 – January 2023
Using the interactive coding environment Dodona in R

BOF starting grant (€210,000)
September 2020 – present
Investigating the added value of machine learning and Twitter data for public health organizations during COVID19.

KCE research grant (€149,132)
April 2022 – April 2024
Budget Impact Projections.

The Data Lab Research Project Grant (£ 19,932.34)
April 2019
Funding for a research project with Actelligent about Sentiment Analysis (together with Dr. Rafaella Calabrese and Prof. Dr. Jonathan Crook)

Grant for participation in a conference abroad (FWO) + grant for participation in a conference (FEB UGent) (€600)
May 2018
Matthias Bogaert, Michel Ballings, Dirk Van den Poel, Asil Oztekin. Comparing the Ability of Facebook and Twitter in Predicting Box Office Sales. 29th European Conference on Operational Research (Valencia, SP).

Grant for participation in a conference abroad (FWO) + grant for participation in a conference (FEB UGent) (€750)
June 2017
Matthias Bogaert, Michel Ballings, Rob Bergmans and Dirk Van den Poel. Predicting Movie Watching Behaviour Using Facebook Data and Information-fusion Sensitivity Analysis. 39th ISMS Marketing Science Conference (Los Angeles, California, USA).

Grant for participation in a conference (FEB UGent) (€750)
November 2015
Matthias Bogaert, Kelly Hewett, Michel Ballings and Dirk Van den Poel. Predicting Buyer Behaviour Using Social Media Data. INFORMS annual meeting 2015 (Philadelphia, Pennsylvania, USA).

MSI Research grant #4-1925 ($4300)
May 2015 – December 2015
Kelly Hewett, Michel Ballings, Matthias Bogaert and Dirk Van den Poel. Harnessing the power of social media to predict buyer behaviour, Marketing Science Institute.

PhD students (current)

Nathan Markx
October 2025 – Present
Title: AGEAS - Automated Generation and Lifecycle Cost Assessment of Manufacturing System Design under Multi-Product Demand Uncertainty

Simon De Lange
November 2024 – Present
Title: Exploring interpretable machine learning models in business analytics.

Wannes Janssens
September 2024 – Present

Title: REMASC - Prediction & Management of the 'Reverse' Remanufacturing Supply Chain.

Yaël De Rocker
September 2024 – Present
Title: REMASC - Prediction & Management of the 'Reverse' Remanufacturing Supply Chain.

Brammert Termont
July 2024 – Present
Title: REMASC - Prediction & Management of the 'Reverse' Remanufacturing Supply Chain.

Xiao Wang
September 2023 – Present
Title: Essays on data analytics in social media

Azin Rashidipour (co-advisor with Prof Dr Broos Maenhout)
September 2023 – Present
Title: Cost-benefit analysis of different types of training for worker reskilling in a production environment

Shimanto Rahman
September 2021 – Present
Title: Applications of Data Analytics in B2B Marketing

PhD students (former)

Lisa Schetgen
April 2019 – September 2022
Title: Social media analytics for good : assessing the value of social media analytics for businesses and public instances

Bram Janssens
September 2018 – September 2022
Title: Essays on interdisciplinary business analytics

Doctoral committees

Juliana Ramirez-Sanchez (IESEG School of Management and Ugent, promotors: Prof Dr Kristof Coussement (IESEG), Prof Dr Arno De Caigny (IESEG), Prof Dr Dries Benoit (Ugent))
February 2026
Title: Applications, challenges, and implementations of AI in B2B business

Arthur Thuy (Ugent, promotor: Prof Dr Dries Benoit)
November 2025
Title: Learning Analytics with Neural Networks: Addressing Open Challenges Through Uncertainty Quantification and Natural Language Processing

Abdel-Jaouad Aberkane (Ugent, promotors: Prof Dr Geert Poels, Prof Dr Seppe vanden Broucke)
June 2025
Title: Automated GDPR Compliance in Requirements Engineering

Benjamin Vermaut (UNamur, promotor: Prof Dr Corentin Burnay)
June 2025

Title: Performance Management in Soccer

Bart De Clerck (Ugent and KMS, promotors: Prof Dr Luis Enrique Correa De Rocha and Prof Dr Ir Filip Van Utterbeeck)
June 2024
Title: Identifying Online Misinformation and Disinformation Spreading Using Graph Theory

Ubaid Mehmood (University of Swinburne, promotor: Prof Dr Irene Moser)
February 2024
Title: Leveraging Social Media Data For More Comprehensive Traffic Load Prediction

Leen De Schaepdrijver (Vlerick Business School, promotor: Prof Dr Philippe Baecke & Prof Dr Koen Tackx)
March 2023
Title: The value of contextual data: how browsing data can advance television advertising

Rafael Van Belle (KULeuven, promotor: Prof Dr Jochen De Weerdt & Prof Dr Bart Baesens)
January 2023
Title: Network Representation Learning for Fraud Detection

Ekaterina Loginova (UGent, promotor: Prof Dr Dries Benoit)
October 2022
Title: Studies on Advanced Natural Language Processing Representations and Deep Learning in Decision Making Systems

Arno Liseune (UGent, promotor: Prof Dr Dirk Van den Poel& Prof Dr Miel Hostens)
March 2022
Title: USING DEEP LEARNING FOR ANIMAL MONITORING TO IMPROVE ANIMAL WELFARE IN DAIRY CATTLE

Bram Steurtewagen (UGent, promotor: Prof Dr Dirk Van den Poel)
December 2021
Title: Predictive Analytics as a value-driver in Industry 4.0

Giselle Van Dongen (UGent, promotor: Prof Dr Dirk Van den Poel)
October 2021
Title: Open Stream Processing Benchmark: an Extensive Analysis of Distributed Stream Processing Frameworks

Refereeing (ad hoc, selected list)

Journal of Service Research
2023 – Present

Information & Management
2023 – Present

Journal of the Operational Research Society
2021 – Present

Omega
2021 – Present

International Journal of Hospitality Management
2020 - Present

Annals of Operations Research
2020 - Present

European Journal of Operational Research
2018 – Present

Journal of Interactive Marketing
2018 - Present

Decision Support Systems
2016 – Present

Service

Adhoc Project Expert Evaluator VLAIO
September 2024 – Present

Senior Editor, Applied Operations and Analytics
September 2024 – Present

Commission Scientific Research, Faculty of Economics and Business Administration, Ghent University
September 2020 – Present

Ethical Commission of Faculty of Economics and Business Administration, Ghent University
September 2022 – September 2023

Follow-up Scientific council Alpha-group, Ghent University
September 2020 – September 2021

Secretary Commission of Master of Data Science for Business
December 2019 – Present

Commission Duurzaamheid (Master of Data Science for Business)
December 2019 – Present

Course Designer Distance Learning at Scale Micromasters
January 2019 – September 2019
Designing a course module (linear regressions) in the online Micromasters in Predictive Analytics

Software

Python package Empulse
Shimanto Rahman, Bram Janssens, and Matthias Bogaert (2024).
Github: https://github.com/ShimantoRahman/empulse
PyPi: https://pypi.org/project/empulse/

Python function B2Boost
Bram Janssens, and Matthias Bogaert (2022).
Github: https://github.com/bram-janssens/B2Boost

R-package DecorateR: Fit and Deploy DECORATE Trees.
Matthias Bogaert, Michel Ballings and Dirk Van den Poel (2017).
Github: https://github.com/MatthBogaert/DecorateR
Cran: https://cran.r-project.org/web/packages/DecorateR/index.html

R-package AggregateR: Aggregate numeric, date and categorical variables by an ID.
Matthias Bogaert, Michel Ballings and Dirk Van den Poel (2015).
Github: https://github.com/MatthBogaert/AggregateR
Cran: https://cran.r-project.org/web/packages/AggregateR/index.html

IT skills

Advanced knowledge:

- R
- Python
- SQL
- Markdown
- MS Office
- Latex

Languages

	Spoken	Written
Dutch	Mother tongue	Mother tongue
English[1]	***	***
French	**	**
German	*	*

[1] Holder of ITACE certificate for CEFR C1-level of Academic English